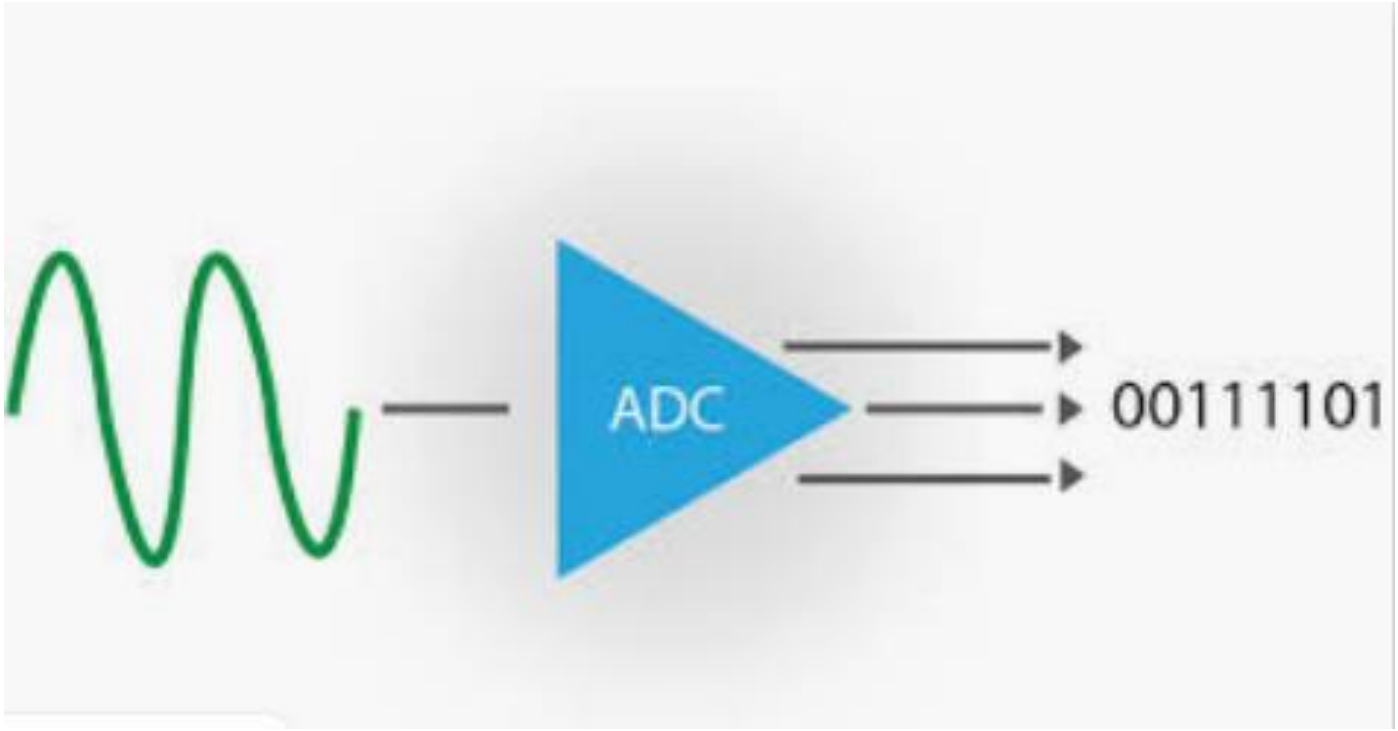
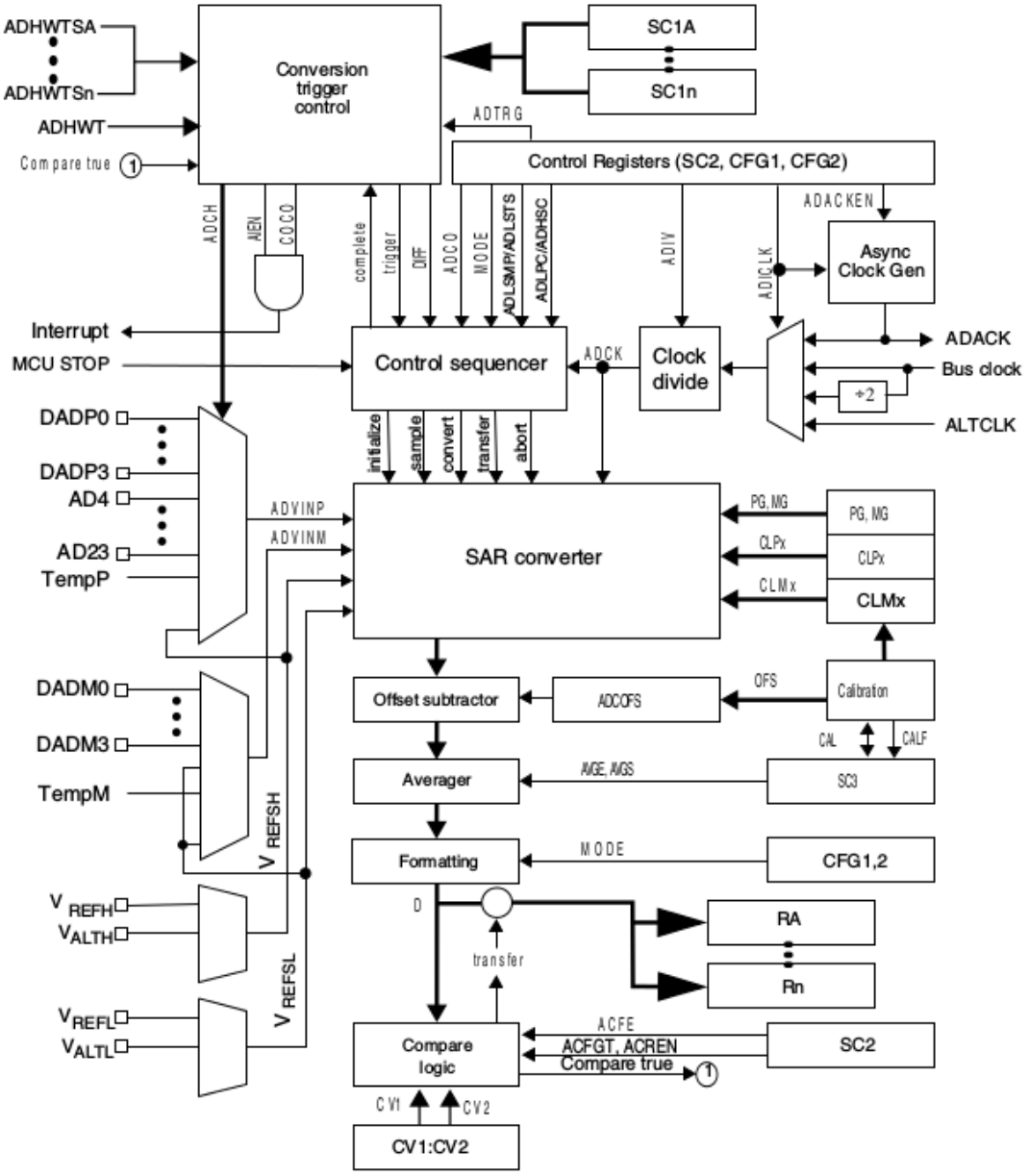
# ADC



# Diagrama en Bloques



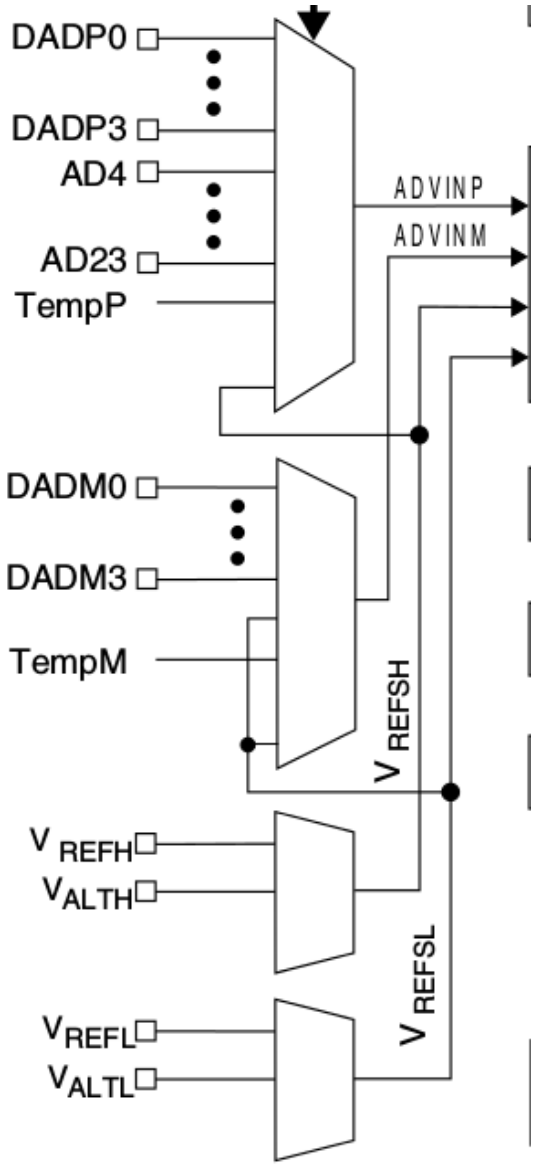
# Registros - Estructura

```
/** ADC - Register Layout Typedef */
typedef struct {
  __IO uint32_t SC1[2];                 /**< ADC Status and Control Registers 1, array offset: 0x0, array step: 0x4 */
  __IO uint32_t CFG1;                   /**< ADC Configuration Register 1, offset: 0x8 */
  __IO uint32_t CFG2;                   /**< ADC Configuration Register 2, offset: 0xC */
  __I  uint32_t R[2];                   /**< ADC Data Result Register, array offset: 0x10, array step: 0x4 */
  __IO uint32_t CV1;                    /**< Compare Value Registers, offset: 0x18 */
  __IO uint32_t CV2;                    /**< Compare Value Registers, offset: 0x1C */
  __IO uint32_t SC2;                    /**< Status and Control Register 2, offset: 0x20 */
  __IO uint32_t SC3;                    /**< Status and Control Register 3, offset: 0x24 */
  __IO uint32_t OFS;                    /**< ADC Offset Correction Register, offset: 0x28 */
  __IO uint32_t PG;                     /**< ADC Plus-Side Gain Register, offset: 0x2C */
  __IO uint32_t MG;                     /**< ADC Minus-Side Gain Register, offset: 0x30 */
  __IO uint32_t CLPD;                   /**< ADC Plus-Side General Calibration Value Register, offset: 0x34 */
  __IO uint32_t CLPS;                   /**< ADC Plus-Side General Calibration Value Register, offset: 0x38 */
  __IO uint32_t CLP4;                   /**< ADC Plus-Side General Calibration Value Register, offset: 0x3C */
  __IO uint32_t CLP3;                   /**< ADC Plus-Side General Calibration Value Register, offset: 0x40 */
  __IO uint32_t CLP2;                   /**< ADC Plus-Side General Calibration Value Register, offset: 0x44 */
  __IO uint32_t CLP1;                   /**< ADC Plus-Side General Calibration Value Register, offset: 0x48 */
  __IO uint32_t CLP0;                   /**< ADC Plus-Side General Calibration Value Register, offset: 0x4C */
       uint8_t RESERVED_0[4];
  __IO uint32_t CLMD;                   /**< ADC Minus-Side General Calibration Value Register, offset: 0x54 */
  __IO uint32_t CLMS;                   /**< ADC Minus-Side General Calibration Value Register, offset: 0x58 */
  __IO uint32_t CLM4;                   /**< ADC Minus-Side General Calibration Value Register, offset: 0x5C */
  __IO uint32_t CLM3;                   /**< ADC Minus-Side General Calibration Value Register, offset: 0x60 */
  __IO uint32_t CLM2;                   /**< ADC Minus-Side General Calibration Value Register, offset: 0x64 */
  __IO uint32_t CLM1;                   /**< ADC Minus-Side General Calibration Value Register, offset: 0x68 */
  __IO uint32_t CLM0;                   /**< ADC Minus-Side General Calibration Value Register, offset: 0x6C */
} ADC_Type, *ADC_MemMapPtr;
```

# Señales de entrada

| Signal | Description |
|---|---|
| DADP3–DADP0 | Differential Analog Channel Inputs |
| DADM3–DADM0 | Differential Analog Channel Inputs |
| AD*n* | Single-Ended Analog Channel Inputs |
| $V_{REFSH}$ | Voltage Reference Select High |
| $V_{REFSL}$ | Voltage Reference Select Low |
| $V_{DDA}$ | Analog Power Supply |
| $V_{SSA}$ | Analog Ground |

- Hasta 24 Canales Single Ended
- 4 Canales Diferenciales



# Señales de entrada

## Table 10-11. ADC 0 Signal Descriptions

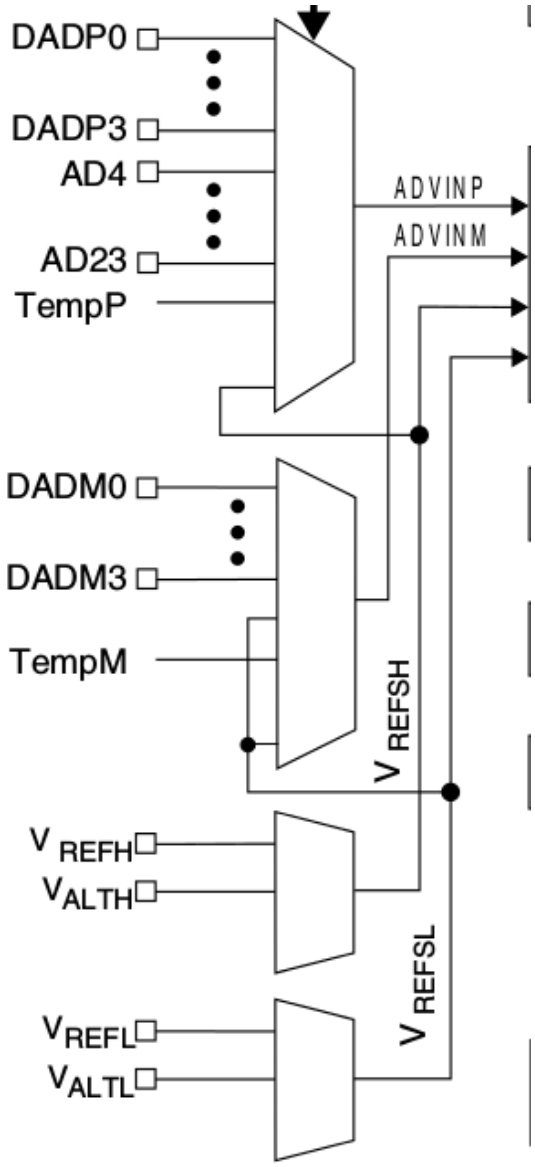
| Chip signal name | Module signal name | Description | I/O |
|---|---|---|---|
| ADC0_DP[3:0] | DADP3–DADP0 | Differential Analog Channel Inputs | I |
| ADC0_DM[3:0] | DADM3–DADM0 | Differential Analog Channel Inputs | I |
| ADC0_SE[18:4] [18,17,15:12,9:4] | AD*n* | Single-Ended Analog Channel Inputs | I |
| VREFH | $V_{REFSH}$ | Voltage Reference Select High | I |
| VREFL | $V_{REFSL}$ | Voltage Reference Select Low | I |
| VDDA | $V_{DDA}$ | Analog Power Supply | I |
| VSSA | $V_{SSA}$ | Analog Ground | I |

## Table 10-12. ADC 1 Signal Descriptions

| Chip signal name | Module signal name | Description | I/O |
|---|---|---|---|
| ADC1_DP3, ADC1_DP[1:0] | DADP3–DADP0 | Differential Analog Channel Inputs | I |
| ADC1_DM3, ADC1_DM[1:0] | DADM3–DADM0 | Differential Analog Channel Inputs | I |
| ADC1_SE[18:418:17, 15:12,9:418:17,15:13 ,9:4] | AD*n* | Single-Ended Analog Channel Inputs | I |
| VREFH | $V_{REFSH}$ | Voltage Reference Select High | I |
| VREFL | $V_{REFSL}$ | Voltage Reference Select Low | I |
| VDDA | $V_{DDA}$ | Analog Power Supply | I |
| VSSA | $V_{SSA}$ | Analog Ground | I |

# Señales de entrada

| Signal | Description |
|---|---|
| DADP3–DADP0 | Differential Analog Channel Inputs |
| DADM3–DADM0 | Differential Analog Channel Inputs |
| AD*n* | Single-Ended Analog Channel Inputs |
| $V_{REFSH}$ | Voltage Reference Select High |
| $V_{REFSL}$ | Voltage Reference Select Low |
| $V_{DDA}$ | Analog Power Supply |
| $V_{SSA}$ | Analog Ground |




- Hasta 24 Canales Single Ended
- 4 Canales Diferenciales

**Notación:**
Diferenciales DAD[x][y]
Single Ended AD[y]

x=M(-) ,P(+)  y= Nro. de Canal

# Selección de Canales e inicio de conversión

El canal a convertir se selecciona con el campo **ADCH[4:0**] del registro **ADCx_SC1**

| | |
|---|---|
| 00000 | When DIFF=0, DADP0 is selected as input; when DIFF=1, DAD0 is selected as input. |
| 00001 | When DIFF=0, DADP1 is selected as input; when DIFF=1, DAD1 is selected as input. |
| 00010 | When DIFF=0, DADP2 is selected as input; when DIFF=1, DAD2 is selected as input. |
| 00011 | When DIFF=0, DADP3 is selected as input; when DIFF=1, DAD3 is selected as input. |
| 00100 | When DIFF=0, AD4 is selected as input; when DIFF=1, it is reserved. |
| 00101 | When DIFF=0, AD5 is selected as input; when DIFF=1, it is reserved. |
| 00110 | When DIFF=0, AD6 is selected as input; when DIFF=1, it is reserved. |
| 00111 | When DIFF=0, AD7 is selected as input; when DIFF=1, it is reserved. |
| 01000 | When DIFF=0, AD8 is selected as input; when DIFF=1, it is reserved. |
| 01001 | When DIFF=0, AD9 is selected as input; when DIFF=1, it is reserved. |
| 01010 | When DIFF=0, AD10 is selected as input; when DIFF=1, it is reserved. |
| 01011 | When DIFF=0, AD11 is selected as input; when DIFF=1, it is reserved. |

**Note:** When DIFF=0 DADP[y] is Single Ended

i.e. DAD**P**[y] → AD[y] y = Nro. de Canal

---

| | |
|---|---|
| 10101 | When DIFF=0, AD21 is selected as input; when DIFF=1, it is reserved. |
| 10110 | When DIFF=0, AD22 is selected as input; when DIFF=1, it is reserved. |
| 10111 | When DIFF=0, AD23 is selected as input; when DIFF=1, it is reserved. |
| 11000 | Reserved. |
| 11001 | Reserved. |
| 11010 | When DIFF=0, Temp Sensor (single-ended) is selected as input; when DIFF=1, Temp Sensor (differential) is selected as input. |
| 11011 | When DIFF=0,Bandgap (single-ended) is selected as input; when DIFF=1, Bandgap (differential) is selected as input. |
| 11100 | Reserved. |
| 11101 | When DIFF=0,$V_{REFSH}$ is selected as input; when DIFF=1, -$V_{REFSH}$ (differential) is selected as input. Voltage reference selected is determined by SC2[REFSEL]. |
| 11110 | When DIFF=0,$V_{REFSL}$ is selected as input; when DIFF=1, it is reserved. Voltage reference selected is determined by SC2[REFSEL]. |
| 11111 | Module is disabled. |

# Asignación de pines **ADC0** para el encapsulado de **100 pines**

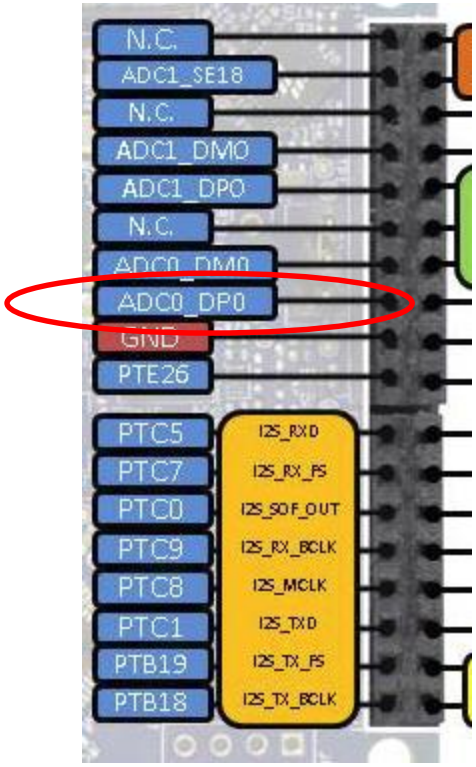
| ADC Channel (SC1n[ADCH]) | Channel | Input signal (SC1n[DIFF]= 1) | Input signal (SC1n[DIFF]= 0) |
|---|---|---|---|
| 00000 | DAD0 | ADC0_DP0 and ADC0_DM0[1] | ADC0_DP0[2] |
| 00001 | DAD1 | ADC0_DP1 and ADC0_DM1 | ADC0_DP1 |
| 00010 | DAD2 | ADC0_DP2 and ADC0_DM2 | ADC0_DP2 |
| 00011 | DAD3 | ADC0_DP3 and ADC0_DM3[3] | ADC0_DP3[4] |
| 00100[5] | AD4a | Reserved | Reserved |
| 00101[5] | AD5a | Reserved | Reserved |
| 00110[5] | AD6a | Reserved | Reserved |
| 00111[5] | AD7a | Reserved | Reserved |
| 00100[5] | AD4b | Reserved | ADC0_SE4b |
| 00101[5] | AD5b | Reserved | ADC0_SE5b |
| 00110[5] | AD6b | Reserved | ADC0_SE6b |
| 00111[5] | AD7b | Reserved | ADC0_SE7b |
| 01000 | AD8 | Reserved | ADC0_SE8[6] |
| 01001 | AD9 | Reserved | ADC0_SE9[7] |
| 01010 | AD10 | Reserved | Reserved |
| 01011 | AD11 | Reserved | Reserved |
| 01100 | AD12 | Reserved | ADC0_SE12 |
| 01101 | AD13 | Reserved | ADC0_SE13 |
| 01110 | AD14 | Reserved | ADC0_SE14 |
| 01111 | AD15 | Reserved | ADC0_SE15 |
| 10000 | AD16 | Reserved | Reserved |
| 10001 | AD17 | Reserved | ADC0_SE17 |
| 10010 | AD18 | Reserved | ADC0_SE18 |
| 10011 | AD19 | Reserved | ADC0_DM0[8] |
| 10100 | AD20 | Reserved | ADC0_DM1 |
| 10101 | AD21 | Reserved | Reserved |
| 10110 | AD22 | Reserved | Reserved |
| 10111 | AD23 | Reserved | 12-bit DAC0 Output/ ADC0_SE23 |
| 11000 | AD24 | Reserved | Reserved |
| 11001 | AD25 | Reserved | Reserved |
| 11010 | AD26 | Temperature Sensor (Diff) | Temperature Sensor (S.E) |
| 11011 | AD27 | Bandgap (Diff)[9] | Bandgap (S.E)[9] |
| 11100 | AD28 | Reserved | Reserved |
| 11101 | AD29 | -VREFH (Diff) | VREFH (S.E) |
| 11110 | AD30 | Reserved | VREFL |
| 11111 | AD31 | Module Disabled | Module Disabled |

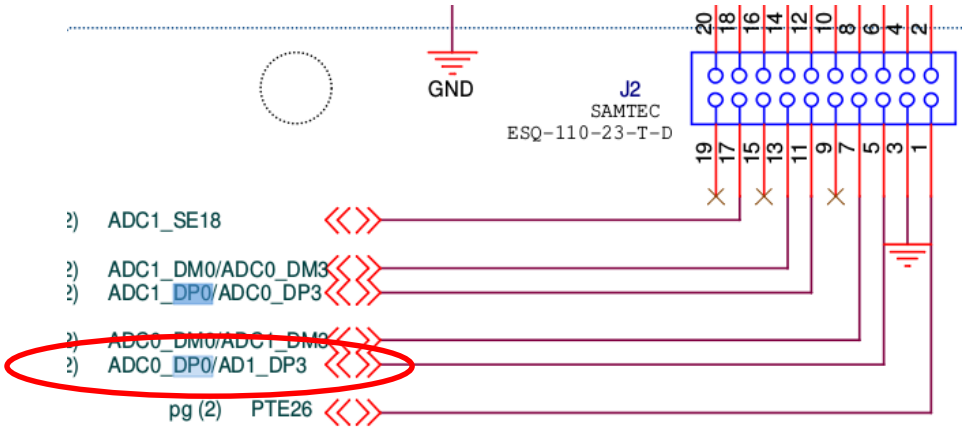
1. Interleaved with ADC1_DP3 and ADC1_DM3
2. Interleaved with ADC1_DP3
3. Interleaved with ADC1_DP0 and ADC1_DM0
4. Interleaved with ADC1_DP0
5. ADCx_CFG2[MUXSEL] bit selects between ADCx_SEn channels a and b. Refer to MUXSEL description in ADC chapter for details.
6. Interleaved with ADC1_SE8
7. Interleaved with ADC1_SE9
8. Interleaved with ADC1_DM3
9. This is the PMC bandgap 1V reference voltage not the VREF module 1.2 V reference voltage. Prior to reading from this ADC channel, ensure that you enable the bandgap buffer by setting the PMC_REGSC[BGBE] bit. Refer to the device data sheet for the bandgap voltage ($V_{BG}$) specification.

# Asignación de pines **ADC1** para el encapsulado de **100 pines**

| ADC Channel (SC1n[ADCH]) | Channel | Input signal (SC1n[DIFF]= 1) | Input signal (SC1n[DIFF]= 0) |
|---|---|---|---|
| 00000 | DAD0 | ADC1_DP0 and ADC1_DM0[1] | ADC1_DP0[2] |
| 00001 | DAD1 | ADC1_DP1 and ADC1_DM1 | ADC1_DP1 |
| 00010 | DAD2 | Reserved | Reserved |
| 00011 | DAD3 | ADC1_DP3 and ADC1_DM3[3] | ADC1_DP3[4] |
| 00100[5] | AD4a | Reserved | ADC1_SE4a |
| 00101[5] | AD5a | Reserved | ADC1_SE5a |
| 00110[5] | AD6a | Reserved | ADC1_SE6a |
| 00111[5] | AD7a | Reserved | ADC1_SE7a |
| 00100[5] | AD4b | Reserved | ADC1_SE4b |
| 00101[5] | AD5b | Reserved | ADC1_SE5b |
| 00110[5] | AD6b | Reserved | ADC1_SE6b |
| 00111[5] | AD7b | Reserved | ADC1_SE7b |
| 01000 | AD8 | Reserved | ADC1_SE8[6] |
| 01001 | AD9 | Reserved | ADC1_SE9[7] |
| 01010 | AD10 | Reserved | Reserved |
| 01011 | AD11 | Reserved | Reserved |
| 01100 | AD12 | Reserved | Reserved |
| 01101 | AD13 | Reserved | Reserved |
| 01110 | AD14 | Reserved | ADC1_SE14 |
| 01111 | AD15 | Reserved | ADC1_SE15 |
| 10000 | AD16 | Reserved | Reserved |
| 10001 | AD17 | Reserved | ADC1_SE17 |
| 10010 | AD18 | Reserved | VREF Output/ADC1_SE18 |
| 10011 | AD19 | Reserved | ADC1_DM0[8] |
| 10100 | AD20 | Reserved | ADC1_DM1 |
| 10101 | AD21 | Reserved | Reserved |
| 10110 | AD22 | Reserved | Reserved |
| 10111 | AD23 | Reserved | Reserved |
| 11000 | AD24 | Reserved | Reserved |
| 11001 | AD25 | Reserved | Reserved |
| 11010 | AD26 | Temperature Sensor (Diff) | Temperature Sensor (S.E) |
| 11011 | AD27 | Bandgap (Diff)[9] | Bandgap (S.E)[9] |
| 11100 | AD28 | Reserved | Reserved |
| 11101 | AD29 | -VREFH (Diff) | VREFH (S.E) |
| 11110 | AD30 | Reserved | VREFL |
| 11111 | AD31 | Module Disabled | Module Disabled |

# Ej: ADC0_DP0 En FRDM-K64F



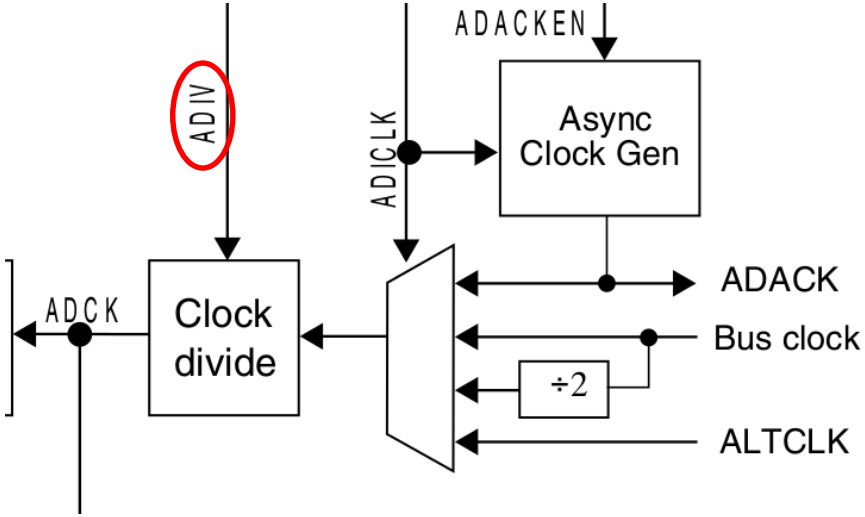

| 144 LQFP | 144 MAP BGA | 121 XFBGA | 100 LQFP | Pin Name | Default | ALT0 |
|---|---|---|---|---|---|---|
| 27 | L1 | K1 | 18 | ADC0_DP0/ ADC1_DP3 | ADC0_DP0/ ADC1_DP3 | ADC0_DP0/ ADC1_DP3 |

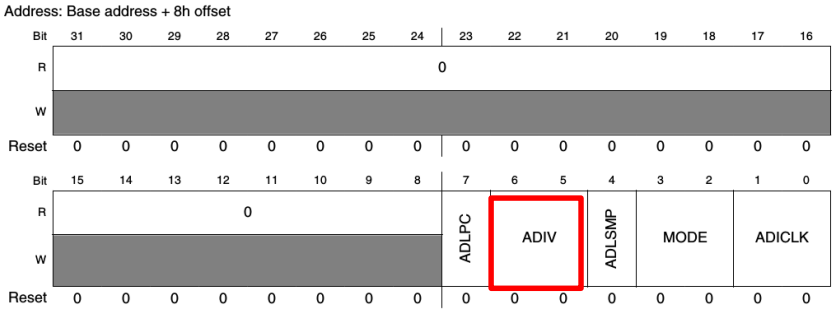
# Clock Para el ADC

El Prescaler para el clock del ADC esta en el registro **ADCx_CFG1[6:5]**

| | |
|---|---|
| 6–5<br>ADIV | Clock Divide Select<br><br>Selects the divide ratio used by the ADC to generate the internal clock ADCK.<br><br>00 The divide ratio is 1 and the clock rate is input clock.<br>01 The divide ratio is 2 and the clock rate is (input clock)/2.<br>10 The divide ratio is 4 and the clock rate is (input clock)/4.<br>11 The divide ratio is 8 and the clock rate is (input clock)/8. |

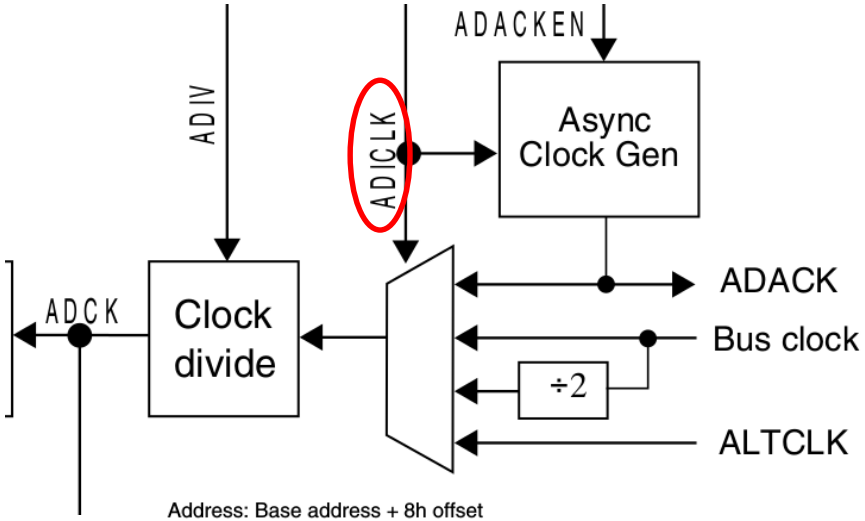
ADC0->CFG1 = ADC_CFG1_ADIV(0x00);



# Clock Para el ADC

El Selector para el clock del ADC esta en el registro **ADCx_CFG1[1:0]**




00 Bus clock
01 Alternate clock 2 (ALTCLK2)
10 Alternate clock (ALTCLK)
11 Asynchronous clock (ADACK)

Address: Base address + 8h offset

| Bit | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R | 0 | | | | | | | | | | | | | | | |
| W | | | | | | | | | | | | | | | | |
| Reset | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| Bit | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R | 0 | | | | | | | | ADLPC | ADIV | | ADLSMP | MODE | | ADICLK | |
| W | | | | | | | | | | | | | | | | |
| Reset | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

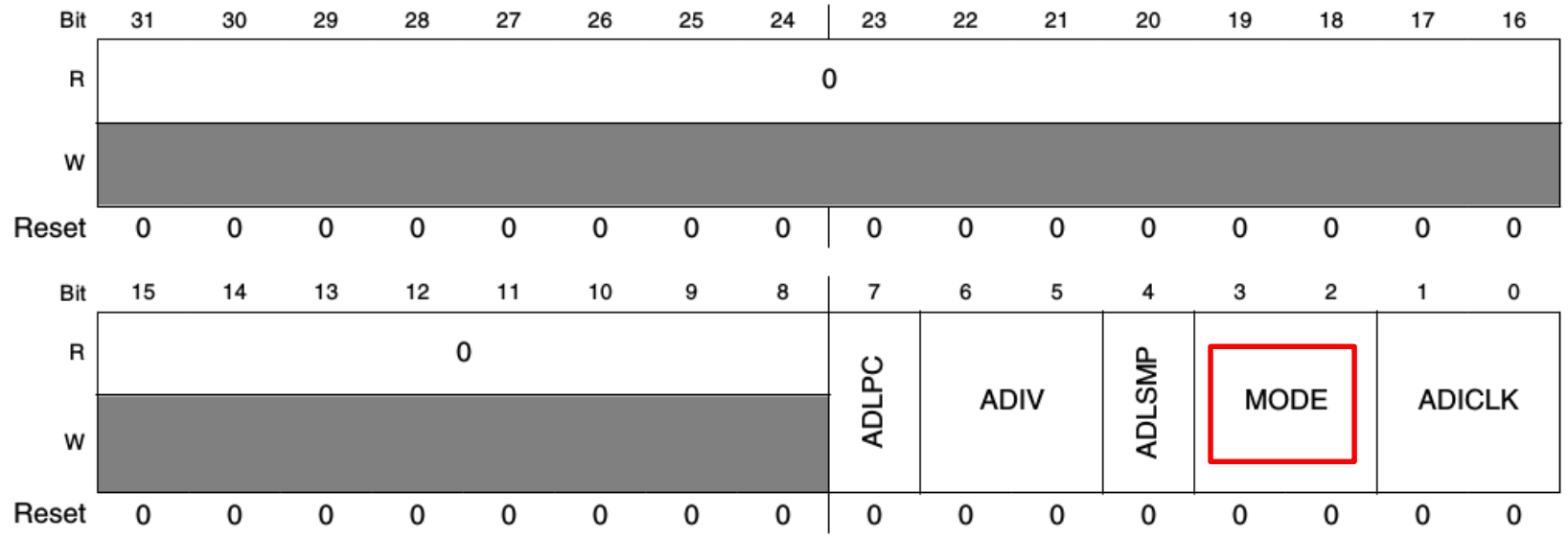
ADC0->CFG1 = ADC_CFG1_ADICLK(0x00) ;

# Resolucion del ADC

La resolución del ADC se configura en el campo **mode** en el registro **ADCx_CFG1[3:2]**

| Mode [3:2] | Diff =0 | Diff=1 (2Complement) |
|---|---|---|
| 00 | 8 bits | 9 bits |
| 01 | 12 bits | 13 bits |
| 10 | 10 bits | 11 bits |
| 11 | 16 bits | 16 bits |

Address: Base address + 8h offset



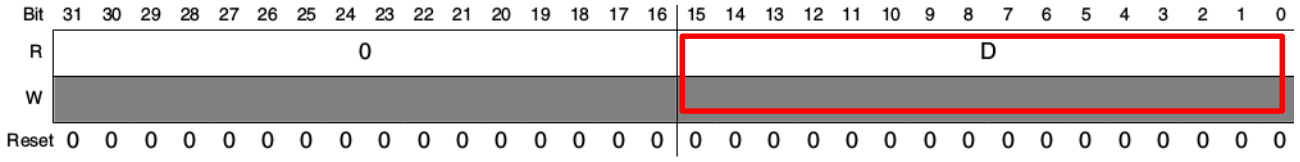
# Data Result Registers del ADC

El Resultado de las conversiones del ADC se encuentra en **ADC Data Result Register (ADCx_Rn)]**

| Conversion mode | D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Format |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 16-bit differential | S | D | D | D | D | D | D | D | D | D | D | D | D | D | D | D | Signed 2's complement |
| 16-bit single-ended | D | D | D | D | D | D | D | D | D | D | D | D | D | D | D | D | Unsigned right justified |
| 13-bit differential | S | S | S | S | D | D | D | D | D | D | D | D | D | D | D | D | Sign-extended 2's complement |
| 12-bit single-ended | 0 | 0 | 0 | 0 | D | D | D | D | D | D | D | D | D | D | D | D | Unsigned right-justified |
| 11-bit differential | S | S | S | S | S | S | D | D | D | D | D | D | D | D | D | D | Sign-extended 2's complement |
| 10-bit single-ended | 0 | 0 | 0 | 0 | 0 | 0 | D | D | D | D | D | D | D | D | D | D | Unsigned right-justified |
| 9-bit differential | S | S | S | S | S | S | S | S | D | D | D | D | D | D | D | D | Sign-extended 2's complement |
| 8-bit single-ended | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | D | D | D | D | D | D | D | D | Unsigned right-justified |

S: Sign bit or sign bit extension;

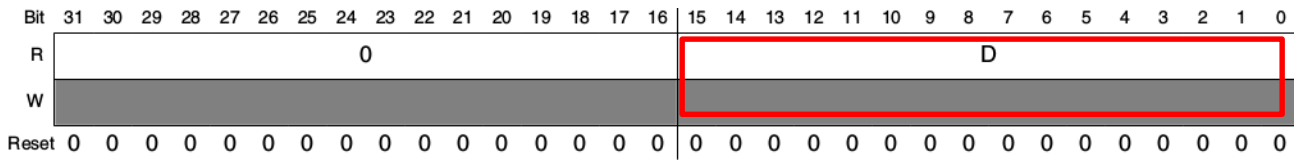
D: Data, which is 2's complement data if indicated




**ADCx_Rn field descriptions**

# Read Data Result Registers del ADC

```
ADCData_t ADC_getData (ADC_t adc)
{
    return adc->R[0];
}
```

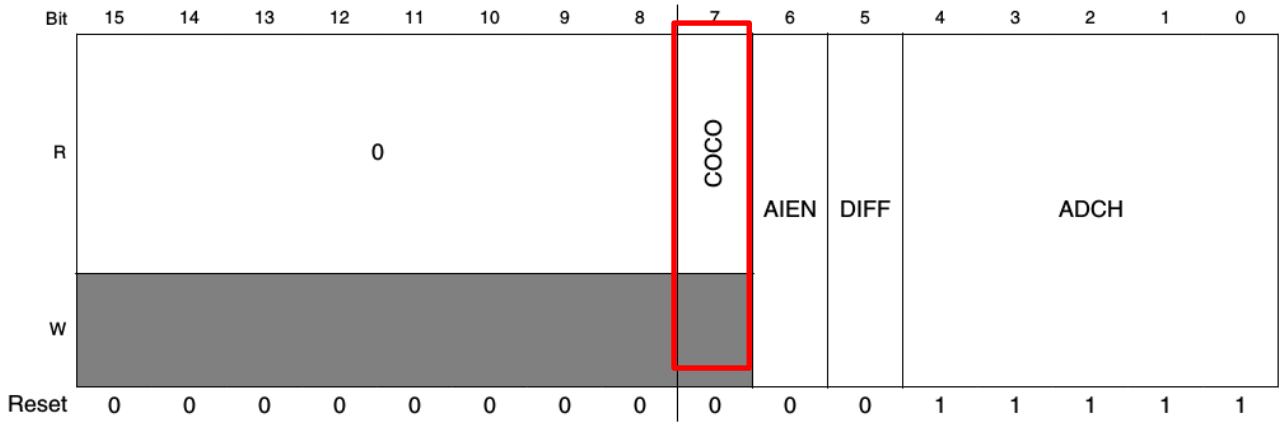



**ADCx_Rn field descriptions**



# EOC del ADC

```
bool ADC_IsReady (ADC_t adc)
{
        return adc->SC1[0] & ADC_SC1_COCO_MASK;
}
```




**ADCx_SC1*n* field descriptions**

# Start of conversion

```
void ADC_Start (ADC_t adc, ADCChannel_t channel, ADCMux_t mux)
{
adc->CFG2 = (adc->CFG2 & ~ADC_CFG2_MUXSEL_MASK) | ADC_CFG2_MUXSEL(mux);

if (adc == ADC0)
adc->SC1[0] = ADC_SC1_AIEN(ADC_interrupt[0]) | ADC_SC1_ADCH(channel);
else if (adc == ADC1)
adc->SC1[0] = ADC_SC1_AIEN(ADC_interrupt[1]) | ADC_SC1_ADCH(channel);
}
```

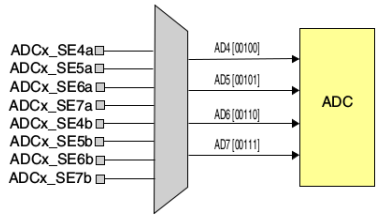
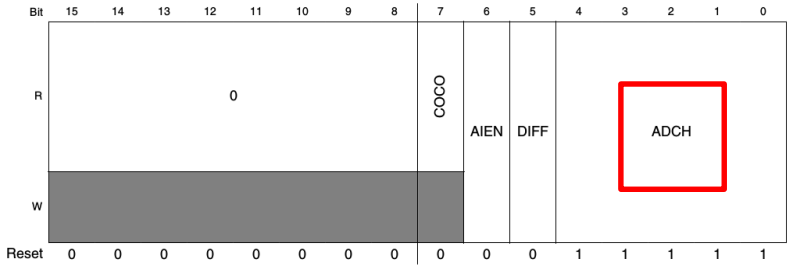



**Figure 3-36. ADCx_SEn channels a and b selection**




**ADCx_SC1*n* field descriptions**